

# Package ‘bfpwr’

October 23, 2025

**Version** 0.2

**Date** 2025-10-23

**Author** Samuel Pawel [aut, cre] (<<https://orcid.org/0000-0003-2779-320X>>)

**Maintainer** Samuel Pawel <samuel.pawel@uzh.ch>

**Title** Power and Sample Size Calculations for Bayes Factor Analysis

**Description** Implements z-test, t-test, and normal moment prior Bayes factors based on summary statistics, along with functionality to perform corresponding power and sample size calculations as described in Pawel and Held (2025) <[doi:10.1080/00031305.2025.2467919](https://doi.org/10.1080/00031305.2025.2467919)>.

**License** GPL-3

**Encoding** UTF-8

**Imports** lamW, mvtnorm, rootSolve

**Suggests** roxygen2, tinytest, knitr

**VignetteBuilder** knitr

**NeedsCompilation** no

**RoxygenNote** 7.3.3

**URL** <https://github.com/SamCH93/bfpwr>

**BugReports** <https://github.com/SamCH93/bfpwr/issues>

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|      |                                       |
|------|---------------------------------------|
| bf01 | <i>Point null z-test Bayes factor</i> |
|------|---------------------------------------|

---

## Description

This function computes the Bayes factor that quantifies the evidence that the data (in the form of an asymptotically normally distributed parameter estimate with standard error) provide for a point null hypothesis with a normal prior assigned to the parameter under the alternative. The standard error is assumed to be known.

## Usage

```
bf01(estimate, se, null = 0, pm, psd, log = FALSE)
```

## Arguments

|          |                                                                                                                                            |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------|
| estimate | Parameter estimate                                                                                                                         |
| se       | Standard error of the parameter estimate                                                                                                   |
| null     | Parameter value under the point null hypothesis. Defaults to 0                                                                             |
| pm       | Mean of the normal prior assigned to the parameter under the alternative                                                                   |
| psd      | Standard deviation of the normal prior assigned to the parameter under the alternative. Set to 0 to obtain a point prior at the prior mean |
| log      | Logical indicating whether the natural logarithm of the Bayes factor should be returned. Defaults to FALSE                                 |

## Value

Bayes factor in favor of the null hypothesis over the alternative ( $BF_{01} > 1$  indicates evidence for the null hypothesis, whereas  $BF_{01} < 1$  indicates evidence for the alternative)

**Author(s)**

Samuel Pawel

**Examples**

```
bf01(estimate = 0.2, se = 0.05, null = 0, pm = 0, psd = 2)
```

binbf01

*Binomial Bayes factor***Description**

This function computes the Bayes factor for testing a binomial proportion  $p$  based on  $x$  observed successes out of  $n$  trials. Two types of tests are available:

- Test of a point null hypothesis: The Bayes factor quantifies the evidence for  $H_0: p = p_0$  against  $H_1: p \neq p_0$ . A beta prior is assigned to the proportion  $p$  under the alternative hypothesis  $H_1$ .
- Test of a directional null hypothesis: The Bayes factor quantifies the evidence for  $H_0: p \leq p_0$  against  $H_1: p > p_0$ . A beta prior that is truncated to the range  $[0, p_0]$  under the null  $H_0$  and to  $(p_0, 1]$  under the alternative  $H_1$  is assigned to the proportion  $p$  under the corresponding hypothesis.

**Usage**

```
binbf01(
  x,
  n,
  p0 = 0.5,
  type = c("point", "direction"),
  a = 1,
  b = 1,
  log = FALSE
)
```

**Arguments**

|                   |                                                                                                            |
|-------------------|------------------------------------------------------------------------------------------------------------|
| <code>x</code>    | Number of successes                                                                                        |
| <code>n</code>    | Number of trials                                                                                           |
| <code>p0</code>   | Tested binomial proportion. Defaults to 0.5                                                                |
| <code>type</code> | Type of test. Can be "point" or "directional". Defaults to "point"                                         |
| <code>a</code>    | Number of successes parameter of the beta prior distribution. Defaults to 1                                |
| <code>b</code>    | Number of failures parameter of the beta prior distribution. Defaults to 1                                 |
| <code>log</code>  | Logical indicating whether the natural logarithm of the Bayes factor should be returned. Defaults to FALSE |

**Value**

Bayes factor in favor of the null hypothesis over the alternative ( $BF_{01} > 1$  indicates evidence for the null hypothesis, whereas  $BF_{01} < 1$  indicates evidence for the alternative)

**Author(s)**

Samuel Pawel

**See Also**

[pbinbf01](#), [nbinbf01](#)

**Examples**

```
## example on Mendelian inheritance from ?stats::binom.test
binbf01(x = 682, n = 925, p0 = 3/4, a = 1, b = 1, type = "point")
## 18.6 => strong evidence for the hypothesized p = 3/4 compared to other p

## with directional hypothesis
binbf01(x = 682, n = 925, p0 = 3/4, a = 1, b = 1, type = "direction")
## 1.5 => only anecdotal evidence for p <= 3/4 over p > 3/4

## Particle-counting experiment from Stone (1997) with point null
binbf01(x = 106298, n = 527135, p0 = 0.2, a = 1, b = 1, type = "point")
## 8.1 => moderate evidence for the alternative over the null

## Coin flip experiment from Bartos et al. (2023) with point null
binbf01(x = 178079, n = 350757, p0 = 0.5, a = 5100, b = 4900, type = "point")
## => 1/1.72e+17 extreme evidence in favor of the alternative over the null
```

---

dirbf01

---

*Directional z-test Bayes factor*


---

**Description**

This function computes the Bayes factor that quantifies the evidence that the data (in the form of an asymptotically normally distributed parameter estimate with standard error) provide for a directional null hypothesis that the parameter value is less than the null value against the alternative that it is greater than the null value. A marginal normal prior is assigned to the parameter. The standard error is assumed to be known.

**Usage**

```
dirbf01(estimate, se, null = 0, pm, psd, log = FALSE)
```

**Arguments**

|          |                                                                                                            |
|----------|------------------------------------------------------------------------------------------------------------|
| estimate | Parameter estimate                                                                                         |
| se       | Standard error of the parameter estimate                                                                   |
| null     | Null value that separates the null from the alternative hypothesis. Defaults to 0                          |
| pm       | Mean of the normal prior assigned to the parameter                                                         |
| psd      | Standard deviation of the normal prior assigned to the parameter                                           |
| log      | Logical indicating whether the natural logarithm of the Bayes factor should be returned. Defaults to FALSE |

**Value**

Bayes factor in favor of the null hypothesis over the alternative ( $BF_{01} > 1$  indicates evidence for the null hypothesis, whereas  $BF_{01} < 1$  indicates evidence for the alternative)

**Author(s)**

Samuel Pawel

**Examples**

```
dirbf01(estimate = 0.2, se = 0.2, null = 0, pm = 0, psd = 2)
```

---

nbf01

---

*Sample size determination for z-test Bayes factor*


---

**Description**

This function computes the required sample size to obtain a Bayes factor ([bf01](#)) more extreme than a threshold  $k$  with a specified target power.

**Usage**

```
nbf01(
  k,
  power,
  usd,
  null = 0,
  pm,
  psd,
  dpm = pm,
  dpsd = psd,
  nrange = c(1, 10^5),
  lower.tail = TRUE,
  integer = TRUE,
  analytical = TRUE,
  ...
)
```

**Arguments**

|                         |                                                                                                                                                                          |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>k</code>          | Bayes factor threshold                                                                                                                                                   |
| <code>power</code>      | Target power                                                                                                                                                             |
| <code>usd</code>        | Unit standard deviation, the (approximate) standard error of the parameter estimate based on $n = 1$ , see details                                                       |
| <code>null</code>       | Parameter value under the point null hypothesis. Defaults to 0                                                                                                           |
| <code>pm</code>         | Mean of the normal prior assigned to the parameter under the alternative in the analysis                                                                                 |
| <code>psd</code>        | Standard deviation of the normal prior assigned to the parameter under the alternative in the analysis. Set to 0 to obtain a point prior at the prior mean               |
| <code>dpm</code>        | Mean of the normal design prior assigned to the parameter. Defaults to the same value as the analysis prior <code>pm</code>                                              |
| <code>dpsd</code>       | Standard deviation of the normal design prior assigned to the parameter. Defaults to the same value as the analysis prior <code>psd</code>                               |
| <code>nrange</code>     | Sample size search range over which numerical search is performed. Defaults to <code>c(1, 10^5)</code>                                                                   |
| <code>lower.tail</code> | Logical indicating whether $\Pr(\text{BF}_{01} \leq k)$ (TRUE) or $\Pr(\text{BF}_{01} > k)$ (FALSE) should be computed. Defaults to TRUE                                 |
| <code>integer</code>    | Logical indicating whether only integer valued sample sizes should be returned. If TRUE the required sample size is rounded to the next larger integer. Defaults to TRUE |
| <code>analytical</code> | Logical indicating whether analytical (if available) or numerical method should be used. Defaults to TRUE                                                                |
| <code>...</code>        | Other arguments passed to <code>stats::uniroot</code>                                                                                                                    |

**Details**

It is assumed that the standard error of the future parameter estimate is of the form  $se = usd/\sqrt{n}$ . For example, for normally distributed data with known standard deviation `sd` and two equally sized groups of size `n`, the standard error of an estimated standardized mean difference is  $se = sd\sqrt{2/n}$ , so the corresponding unit standard deviation is  $usd = sd\sqrt{2}$ . See the vignette for more information.

**Value**

The required sample size to achieve the specified power

**Note**

A warning message will be displayed in case that the specified target power is not achievable under the specified analysis and design priors.

**Author(s)**

Samuel Pawel

**See Also**[pbf01](#), [powerbf01](#), [bf01](#)**Examples**

```
## point alternative (analytical and numerical solution available)
nbf01(k = 1/10, power = 0.9, usd = 1, null = 0, pm = 0.5, psd = 0,
      analytical = c(TRUE, FALSE), integer = FALSE)

## standardized mean difference (usd = sqrt(2), effective sample size = per group size)
nbf01(k = 1/10, power = 0.9, usd = sqrt(2), null = 0, pm = 0, psd = 1)
## this is the sample size per group (assuming equally sized groups)

## z-transformed correlation (usd = 1, effective sample size = n - 3)
nbf01(k = 1/10, power = 0.9, usd = 1, null = 0, pm = 0.2, psd = 0.5)
## have to add 3 to obtain the actual sample size

## log hazard/odds ratio (usd = 2, effective sample size = total number of events)
nbf01(k = 1/10, power = 0.9, usd = 2, null = 0, pm = 0, psd = sqrt(0.5))
## have to convert the number of events to a sample size
```

nbinbf01

*Sample size determination for binomial Bayes factor***Description**

This function computes the required sample size to obtain a binomial Bayes factor ([binbf01](#)) more extreme than a threshold  $k$  with a specified target power.

**Usage**

```
nbinbf01(
  k,
  power,
  p0 = 0.5,
  type = c("point", "direction"),
  a = 1,
  b = 1,
  dp = NA,
  da = a,
  db = b,
  dl = 0,
  du = 1,
  lower.tail = TRUE,
  nrange = c(1, 10^4),
  ...
)
```

**Arguments**

|            |                                                                                                                                                                                        |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| k          | Bayes factor threshold                                                                                                                                                                 |
| power      | Target power                                                                                                                                                                           |
| p0         | Tested binomial proportion. Defaults to 0.5                                                                                                                                            |
| type       | Type of test. Can be "point" or "directional". Defaults to "point"                                                                                                                     |
| a          | Number of successes parameter of the beta analysis prior distribution. Defaults to 1                                                                                                   |
| b          | Number of failures parameter of the beta analysis prior distribution. Defaults to 1                                                                                                    |
| dp         | Fixed binomial proportion assumed for the power calculation. Set to NA to use a truncated beta design prior instead (specified via the da, db, dl, and du arguments). Defaults to NA   |
| da         | Number of successes parameter of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to the same value a as specified for the analysis prior |
| db         | Number of failures parameter of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to the same value b as specified for the analysis prior  |
| dl         | Lower truncation limit of of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to 0                                                        |
| du         | Upper truncation limit of of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to 1                                                        |
| lower.tail | Logical indicating whether $\Pr(\text{BF}_{01} \leq k)$ (TRUE) or $\Pr(\text{BF}_{01} > k)$ (FALSE) should be computed. Defaults to TRUE                                               |
| nrange     | Sample size search range over which numerical search is performed. Defaults to $c(1, 10^4)$                                                                                            |
| ...        | Other arguments passed to <code>stats::uniroot</code>                                                                                                                                  |

**Value**

The required sample size to achieve the specified power

**Author(s)**

Samuel Pawel

**See Also**

[pbinbf01](#), [binbf01](#)



## Examples

```
## sample size parameters
pow <- 0.9
p0 <- 3/4
a <- 1
b <- 1
k <- 1/10

## Not run:
## sample sizes for directional testing
(nH1 <- nbinbf01(k = k, power = pow, p0 = p0, type = "direction", a = a,
  b = b, da = a, db = b, dl = p0, du = 1))
(nH0 <- nbinbf01(k = 1/k, power = pow, p0 = p0, type = "direction", a = a,
  b = b, da = a, db = b, dl = 0, du = p0, lower.tail = FALSE))
nseq <- seq(1, 1.1*max(c(nH1, nH0)), length.out = 100)
powH1 <- pbinbf01(k = k, n = nseq, p0 = p0, type = "direction", a = a,
  b = b, da = a, db = b, dl = p0, du = 1)
powH0 <- pbinbf01(k = 1/k, n = nseq, p0 = p0, type = "direction", a = a,
  b = b, da = a, db = b, dl = 0, du = p0, lower.tail = FALSE)
matplot(nseq, cbind(powH1, powH0), type = "s", xlab = "n", ylab = "Power", lty = 1,
  ylim = c(0, 1), col = c(2, 4), las = 1)
abline(h = pow, lty = 2)
abline(v = c(nH1, nH0), col = c(2, 4), lty = 2)
legend("topleft", legend = c("H1", "H0"), lty = 1, col = c(2, 4))

## sample sizes for point null testing
(nH1 <- nbinbf01(k = k, power = pow, p0 = p0, type = "point", a = a,
  b = b, da = a, db = b))
(nH0 <- nbinbf01(k = 1/k, power = pow, p0 = p0, type = "point", a = a,
  b = b, dp = p0, lower.tail = FALSE, nrange = c(1, 10^5)))
nseq <- seq(1, max(c(nH1, nH0)), length.out = 100)
powH1 <- pbinbf01(k = k, n = nseq, p0 = p0, type = "point", a = a,
  b = b, da = a, db = b, dl = 0, du = 1)
powH0 <- pbinbf01(k = 1/k, n = nseq, p0 = p0, type = "point", a = a,
  b = b, dp = p0, lower.tail = FALSE)
matplot(nseq, cbind(powH1, powH0), type = "s", xlab = "n", ylab = "Power", lty = 1,
  ylim = c(0, 1), col = c(2, 4), las = 1)
abline(h = pow, lty = 2)
abline(v = c(nH1, nH0), col = c(2, 4), lty = 2)
legend("topleft", legend = c("H1", "H0"), lty = 1, col = c(2, 4))

## End(Not run)
```

---

nmbf01

*Normal moment prior Bayes factor*


---

## Description

This function computes the Bayes factor that quantifies the evidence that the data (in the form of an asymptotically normally distributed parameter estimate with standard error) provide for a point null hypothesis with a normal moment prior assigned to the parameter under the alternative.

**Usage**

```
nmbf01(estimate, se, null = 0, psd, log = FALSE)
```

**Arguments**

|          |                                                                                                                                            |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------|
| estimate | Parameter estimate                                                                                                                         |
| se       | Standard error of the parameter estimate                                                                                                   |
| null     | Parameter value under the point null hypothesis. Defaults to 0                                                                             |
| psd      | Spread of the normal moment prior assigned to the parameter under the alternative. The modes of the prior are located at $\pm\sqrt{2}$ psd |
| log      | Logical indicating whether the natural logarithm of the Bayes factor should be returned. Defaults to FALSE                                 |

**Details**

A normal moment prior has density  $f(x \mid \text{null}, \text{psd}) = N(x \mid \text{null}, \text{psd}^2) \times (x - \text{null})/\text{psd}^2$  with  $N(x \mid m, v)$  the normal density with mean  $m$  and variance  $v$  evaluated at  $x$ .

**Value**

Bayes factor in favor of the null hypothesis over the alternative ( $\text{BF}_{01} > 1$  indicates evidence for the null hypothesis, whereas  $\text{BF}_{01} < 1$  indicates evidence for the alternative)

**Author(s)**

Samuel Pawel

**References**

- Johnson, V. E. and Rossell, D. (2010). On the use of non-local prior densities in Bayesian hypothesis tests. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 72(2):143–170. [doi:10.1111/j.14679868.2009.00730.x](https://doi.org/10.1111/j.14679868.2009.00730.x)
- Pramanik, S. and Johnson, V. E. (2024). Efficient alternatives for Bayesian hypothesis tests in psychology. *Psychological Methods*, 29(2):243–261. [doi:10.1037/met0000482](https://doi.org/10.1037/met0000482)

**See Also**

[nmbf01](#), [pnmbf01](#), [nnmbf01](#), [powernmbf01](#)

**Examples**

```
nmbf01(estimate = 0.25, se = 0.05, null = 0, psd = 0.5/sqrt(2)) # mode at 0.5
```

nnmbf01

*Sample size determination for normal moment prior Bayes factor***Description**

This function computes the required sample size to obtain a normal moment prior Bayes factor ([nbf01](#)) more extreme than a threshold  $k$  with a specified target power.

**Usage**

```
nnmbf01(
  k,
  power,
  usd,
  null = 0,
  psd,
  dpm,
  dpsd,
  nrange = c(1, 10^5),
  lower.tail = TRUE,
  integer = TRUE,
  ...
)
```

**Arguments**

|                         |                                                                                                                                                                          |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>k</code>          | Bayes factor threshold                                                                                                                                                   |
| <code>power</code>      | Target power                                                                                                                                                             |
| <code>usd</code>        | Unit standard deviation, the (approximate) standard error of the parameter estimate based on $n = 1$ , see details                                                       |
| <code>null</code>       | Parameter value under the point null hypothesis. Defaults to 0                                                                                                           |
| <code>psd</code>        | Spread of the normal moment prior assigned to the parameter under the alternative in the analysis. The modes of the prior are located at $\pm\sqrt{2}$ psd               |
| <code>dpm</code>        | Mean of the normal design prior assigned to the parameter                                                                                                                |
| <code>dpsd</code>       | Standard deviation of the normal design prior assigned to the parameter. Set to 0 to obtain a point prior at the design prior mean                                       |
| <code>nrange</code>     | Sample size search range over which numerical search is performed. Defaults to $c(1, 10^5)$                                                                              |
| <code>lower.tail</code> | Logical indicating whether $\Pr(\text{BF}_{01} \leq k)$ (TRUE) or $\Pr(\text{BF}_{01} > k)$ (FALSE) should be computed. Defaults to TRUE                                 |
| <code>integer</code>    | Logical indicating whether only integer valued sample sizes should be returned. If TRUE the required sample size is rounded to the next larger integer. Defaults to TRUE |
| <code>...</code>        | Other arguments passed to <code>stats::uniroot</code>                                                                                                                    |

### Details

It is assumed that the standard error of the future parameter estimate is of the form  $se = usd/\sqrt{n}$ . For example, for normally distributed data with known standard deviation  $sd$  and two equally sized groups of size  $n$ , the standard error of an estimated standardized mean difference is  $se = sd\sqrt{2/n}$ , so the corresponding unit standard deviation is  $usd = sd\sqrt{2}$ . See the vignette for more information.

### Value

The required sample size to achieve the specified power

### Author(s)

Samuel Pawel

### See Also

[nmbf01](#), [pnmbf01](#), [powernmbf01](#)

### Examples

```
nnmbf01(k = 1/10, power = 0.9, usd = 1, null = 0, psd = 0.5/sqrt(2), dpm = 0.5, dpsd = 0)
```

---

ntbf01

---

*Sample size calculations for t-test Bayes factor*


---

### Description

This function computes the required sample size to obtain a  $t$ -test Bayes factor ([tbf01](#)) more extreme than a threshold  $k$  with a specified target power.

### Usage

```
ntbf01(
  k,
  power,
  null = 0,
  plocation = 0,
  pscale = 1/sqrt(2),
  pdf = 1,
  type = c("two.sample", "one.sample", "paired"),
  alternative = c("two.sided", "less", "greater"),
  dpm = plocation,
  dpsd = pscale,
  lower.tail = TRUE,
  integer = TRUE,
  nrange = c(2, 10^4),
  ...
)
```

**Arguments**

|                          |                                                                                                                                                                                                                                                                                                                |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>k</code>           | Bayes factor threshold                                                                                                                                                                                                                                                                                         |
| <code>power</code>       | Target power                                                                                                                                                                                                                                                                                                   |
| <code>null</code>        | Standardized mean difference under the point null hypothesis. Defaults to 0                                                                                                                                                                                                                                    |
| <code>plocation</code>   | $t$ prior location. Defaults to 0                                                                                                                                                                                                                                                                              |
| <code>pscale</code>      | $t$ prior scale. Defaults to $1/\sqrt{2}$                                                                                                                                                                                                                                                                      |
| <code>pdf</code>         | $t$ prior degrees of freedom. Defaults to 1 (a Cauchy prior)                                                                                                                                                                                                                                                   |
| <code>type</code>        | Type of $t$ -test. Can be "two.sample" (default), "one.sample", or "paired"                                                                                                                                                                                                                                    |
| <code>alternative</code> | Direction of the test. Can be either "two.sided" (default), "less", or "greater". The latter two truncate the analysis prior to negative and positive effects, respectively. If set to "less" or "greater", the power is only computed based on data with effect estimates in the direction of the alternative |
| <code>dpm</code>         | Mean of the normal design prior assigned to the standardized mean difference. Defaults to the analysis prior location                                                                                                                                                                                          |
| <code>dpsd</code>        | Standard deviation of the normal design prior assigned to the standardized mean difference. Set to 0 to obtain a point prior at the design prior mean. Defaults to the analysis prior scale                                                                                                                    |
| <code>lower.tail</code>  | Logical indicating whether $\Pr(\text{BF}_{01} \leq k)$ (TRUE) or $\Pr(\text{BF}_{01} > k)$ (FALSE) should be computed. Defaults to TRUE                                                                                                                                                                       |
| <code>integer</code>     | Logical indicating whether only integer valued sample sizes should be returned. If TRUE the required sample size is rounded to the next larger integer. Defaults to TRUE                                                                                                                                       |
| <code>nrange</code>      | Sample size search range over which numerical search is performed. Defaults to <code>c(2, 10^4)</code>                                                                                                                                                                                                         |
| <code>...</code>         | Other arguments passed to <code>stats::uniroot</code>                                                                                                                                                                                                                                                          |

**Value**

The required sample size to achieve the specified power

**Author(s)**

Samuel Pawel

**See Also**

[ptbf01](#), [powertbf01](#), [tbf01](#)

**Examples**

```
## example from Schönbrodt and Wagenmakers (2018, p.135)
ntbf01(k = 1/6, power = 0.95, dpm = 0.5, dpsd = 0, alternative = "greater")
ntbf01(k = 1/6, power = 0.95, dpm = 0.5, dpsd = 0.1, alternative = "greater")
ntbf01(k = 6, power = 0.95, dpm = 0, dpsd = 0, alternative = "greater",
      lower.tail = FALSE, nrange = c(2, 10000))
```

pbf01

*Cumulative distribution function of the z-test Bayes factor***Description**

This function computes the probability of obtaining a Bayes factor ([bfb01](#)) more extreme than a threshold  $k$  with a specified sample size.

**Usage**

```
pbf01(k, n, usd, null = 0, pm, psd, dpm = pm, dpsd = psd, lower.tail = TRUE)
```

**Arguments**

|                         |                                                                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>k</code>          | Bayes factor threshold                                                                                                                                     |
| <code>n</code>          | Sample size                                                                                                                                                |
| <code>usd</code>        | Unit standard deviation, the (approximate) standard error of the parameter estimate based on $n = 1$ , see details                                         |
| <code>null</code>       | Parameter value under the point null hypothesis. Defaults to 0                                                                                             |
| <code>pm</code>         | Mean of the normal prior assigned to the parameter under the alternative in the analysis                                                                   |
| <code>psd</code>        | Standard deviation of the normal prior assigned to the parameter under the alternative in the analysis. Set to 0 to obtain a point prior at the prior mean |
| <code>dpm</code>        | Mean of the normal design prior assigned to the parameter. Defaults to the same value as the analysis prior <code>pm</code>                                |
| <code>dpsd</code>       | Standard deviation of the normal design prior assigned to the parameter. Defaults to the same value as the analysis prior <code>psd</code>                 |
| <code>lower.tail</code> | Logical indicating whether $\Pr(BF_{01} \leq k)$ (TRUE) or $\Pr(BF_{01} > k)$ (FALSE) should be computed. Defaults to TRUE                                 |

**Details**

It is assumed that the standard error of the future parameter estimate is of the form  $se = usd/\sqrt{n}$ . For example, for normally distributed data with known standard deviation `sd` and two equally sized groups of size `n`, the standard error of an estimated standardized mean difference is  $se = sd\sqrt{2/n}$ , so the corresponding unit standard deviation is  $usd = sd\sqrt{2}$ . See the vignette for more information.

**Value**

The probability that the Bayes factor is less or greater (depending on the specified `lower.tail`) than the specified threshold `k`

**Author(s)**

Samuel Pawel

**See Also**

[nbf01](#), [powerbf01](#), [bf01](#)

**Examples**

```
## point alternative (psd = 0)
pbf01(k = 1/10, n = 200, usd = 2, null = 0, pm = 0.5, psd = 0)

## normal alternative (psd > 0)
pbf01(k = 1/10, n = 100, usd = 2, null = 0, pm = 0.5, psd = 2)

## design prior is the null hypothesis (dpm = 0, dpsd = 0)
pbf01(k = 10, n = 1000, usd = 2, null = 0, pm = 0.3, psd = 2, dpm = 0, dpsd = 0, lower.tail = FALSE)

## draw a power curve
nseq <- round(exp(seq(log(10), log(10000), length.out = 100)))
plot(nseq, pbf01(k = 1/10, n = nseq, usd = 2, null = 0, pm = 0.3, psd = 0), type = "l",
     xlab = "n", ylab = bquote("Pr(BF\"01\" <= 1/10 * ")"), ylim = c(0, 1),
     log = "x", las = 1)

## standardized mean difference (usd = sqrt(2), effective sample size = per group size)
n <- 30
pbf01(k = 1/10, n = n, usd = sqrt(2), null = 0, pm = 0, psd = 1)

## z-transformed correlation (usd = 1, effective sample size = n - 3)
n <- 100
pbf01(k = 1/10, n = n - 3, usd = 1, null = 0, pm = 0.2, psd = 0.5)

## log hazard/odds ratio (usd = 2, effective sample size = total number of events)
nevents <- 100
pbf01(k = 1/10, n = nevents, usd = 2, null = 0, pm = 0, psd = sqrt(0.5))
```

---

pbf01seq

---

*Sequential Bayes Factor Design*


---

**Description**

Computes cumulative probabilities of observing  $z$ -test Bayes factors that provide evidence for the null hypothesis  $H_0$ , the alternative hypothesis  $H_1$ , or remain inconclusive in a sequential design. Optionally, also computes the expected sample size.

**Usage**

```
pbf01seq(
  k1,
  k0 = 1/k1,
  se,
  n = NULL,
```

```

    pm = NULL,
    psd,
    dpm = pm,
    dpsd = psd,
    type = c("normal", "directional", "moment"),
    strict = TRUE,
    ...
)

```

### Arguments

|        |                                                                                                                                                                                                                                                                                                                                                                       |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| k1     | Bayes factor threshold in favor of $H_1$ (i.e., $BF_{01} \leq k1 < 1$ implies evidence for $H_1$ )                                                                                                                                                                                                                                                                    |
| k0     | Bayes factor threshold in favor of $H_0$ (i.e., $BF_{01} \geq k0 > 1$ implies evidence for $H_1$ )                                                                                                                                                                                                                                                                    |
| se     | Numeric vector of standard errors for each sequential stage                                                                                                                                                                                                                                                                                                           |
| n      | Optional numeric vector of sample sizes corresponding to se. If supplied, the expected sample size is computed                                                                                                                                                                                                                                                        |
| pm     | Analysis prior mean. Not taken into account for type = "moment"                                                                                                                                                                                                                                                                                                       |
| psd    | Analysis prior standard deviation (type = "moment" and type = "directional") or scale (type = "moment")                                                                                                                                                                                                                                                               |
| dpm    | Mean of the normal design prior                                                                                                                                                                                                                                                                                                                                       |
| dpsd   | Standard deviation of the normal design prior. Set dpsd = 0 to obtain a point prior at dpm                                                                                                                                                                                                                                                                            |
| type   | Character string. One of <ul style="list-style-type: none"> <li>"normal" (default): point null vs. normal alternative (set psd = 0 to obtain a point alternative)</li> <li>"directional": directional null vs. directional alternative with a marginal normal prior</li> <li>"moment": point null vs. normal moment alternative which is centered around 0</li> </ul> |
| strict | Logical. If TRUE and there are more than two critical values per stage, integrate over all possible region combinations (slow but exact). If FALSE, only integrates over the main regions where the sign of the z-statistics does not change across stages (faster, recommended when many interim analyses, e.g., more than 10, are performed). Defaults to TRUE      |
| ...    | Additional arguments passed to <code>mvtnorm::lpmvnorm</code>                                                                                                                                                                                                                                                                                                         |

### Details

The function constructs per-stage integration regions for cumulative z-statistics based on the Bayes factor thresholds k1 and k0, then computes the probability of these regions under a predictive distribution defined by se and the normal design prior with dpm and dpsd. Integration is performed via `mvtnorm::lpmvnorm`.



**Value**

An object of class "bfseqdesign", which is a list containing the input arguments, the critical z-values, the expected sample size, the cumulative probabilities of stopping for  $H_1$  and  $H_0$  by each stage, and the cumulative probabilities of remaining inconclusive by each stage.

**Author(s)**

Samuel Pawel

**Examples**

```
n <- seq(50, 200, 50) # sample size per stage
se <- sqrt(2/n) # standard errors per stage
res <- pbf01seq(k1 = 1/10, k0 = 3, se = se, n = n, pm = 0, psd = 1,
               dpm = 0.5, dpsd = 0.05, type = "normal")
res # print summary
plot(res) # plot summary
res$cumpH1 # cumulative probability to stop for H1 by each stage
res$cumpH0 # cumulative probability to stop for H0 by each stage
res$EN # expected sample size
```

---

pbinbf01

---

*Cumulative distribution function of the binomial Bayes factor*


---

**Description**

This function computes the probability of obtaining a binomial Bayes factor ([binbf01](#)) more extreme than a threshold  $k$  with a specified sample size.

**Usage**

```
pbinbf01(
  k,
  n,
  p0 = 0.5,
  type = c("point", "direction"),
  a = 1,
  b = 1,
  dp = NA,
  da = a,
  db = b,
  dl = 0,
  du = 1,
  lower.tail = TRUE
)
```

**Arguments**

|            |                                                                                                                                                                                        |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| k          | Bayes factor threshold                                                                                                                                                                 |
| n          | Number of trials                                                                                                                                                                       |
| p0         | Tested binomial proportion. Defaults to 0.5                                                                                                                                            |
| type       | Type of test. Can be "point" or "directional". Defaults to "point"                                                                                                                     |
| a          | Number of successes parameter of the beta analysis prior distribution. Defaults to 1                                                                                                   |
| b          | Number of failures parameter of the beta analysis prior distribution. Defaults to 1                                                                                                    |
| dp         | Fixed binomial proportion assumed for the power calculation. Set to NA to use a truncated beta design prior instead (specified via the da, db, dl, and du arguments). Defaults to NA   |
| da         | Number of successes parameter of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to the same value a as specified for the analysis prior |
| db         | Number of failures parameter of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to the same value b as specified for the analysis prior  |
| dl         | Lower truncation limit of of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to 0                                                        |
| du         | Upper truncation limit of of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to 1                                                        |
| lower.tail | Logical indicating whether $\Pr(\text{BF}_{01} \leq k)$ (TRUE) or $\Pr(\text{BF}_{01} > k)$ (FALSE) should be computed. Defaults to TRUE                                               |

**Value**

The probability that the Bayes factor is less or greater (depending on the specified lower.tail) than the specified threshold k

**Author(s)**

Samuel Pawel

**See Also**

[binbf01](#), [nbinbf01](#)

**Examples**

```
## compute probability that BF > 10 under the point null
a <- 1
b <- 1
p0 <- 3/4
k <- 10
nseq <- seq(1, 1000, length.out = 100)
```

```

powH0 <- pbinbf01(k = k, n = nseq, p0 = p0, type = "point", a = a, b = b,
                  dp = p0, lower.tail = FALSE)
plot(nseq, powH0, type = "s", xlab = "n", ylab = "Power")

## compare to normal approximation
pm <- a/(a + b) # prior mean under H1
psd <- sqrt(a*b/(a + b)^2/(a + b + 1)) # prior standard deviation under H1
pownormH0 <- pbf01(k = k, n = nseq, usd = sqrt(p0*(1 - p0)), null = p0,
                  pm = pm, psd = psd, dpm = p0, dpsd = 0, lower.tail = FALSE)
lines(nseq, pownormH0, type = "s", col = 2)
legend("right", legend = c("Exact", "Normal approximation"), lty = 1,
      col = c(1, 2))

## compute probability that BF < 1/10 under the p|H1 ~ Beta(a, b) alternative
a <- 10
b <- 5
p0 <- 3/4
k <- 1/10
powH1 <- pbinbf01(k = k, n = nseq, p0 = p0, type = "point", a = a, b = b,
                  da = a, db = b, dl = 0, du = 1)
plot(nseq, powH1, type = "s", xlab = "n", ylab = "Power")

## compare to normal approximation
pm <- a/(a + b) # prior mean under H1
psd <- sqrt(a*b/(a + b)^2/(a + b + 1)) # prior standard deviation under H1
pownormH1 <- pbf01(k = k, n = nseq, usd = sqrt(pm*(1 - pm)), null = p0,
                  pm = pm, psd = psd, dpm = pm, dpsd = psd)
lines(nseq, pownormH1, type = "s", col = 2)
legend("right", legend = c("Exact", "Normal approximation"), lty = 1,
      col = c(1, 2))

## probability that directional BF <= 1/10 under uniform [3/4, 1] design prior
pow <- pbinbf01(k = 1/10, n = nseq, p0 = 3/4, type = "direction", a = 1, b = 1,
                da = 1, db = 1, dl = 3/4, du = 1)
plot(nseq, pow, type = "s", xlab = "n", ylab = "Power")

```

---

plot.bfseqdesign

*Plot method for class "bfseqdesign"*


---

## Description

Plot method for class "bfseqdesign"

## Usage

```

## S3 method for class 'bfseqdesign'
plot(
  x,
  plot = TRUE,
  nullplot = TRUE,

```

```

    zplot = FALSE,
    digits = max(3L, getOption("digits") - 3L),
    ...
  )

```

### Arguments

|                       |                                                                                                                                                       |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>x</code>        | Object of class "power.bftest"                                                                                                                        |
| <code>plot</code>     | Logical indicating whether data should be plotted. If FALSE only the data used for plotting are returned.                                             |
| <code>nullplot</code> | Logical indicating whether a second plot with the stopping probabilities computed under the null hypothesis should also be produced. Defaults to TRUE |
| <code>zplot</code>    | Logical indicating whether a plot of the critical z-values should be produced                                                                         |
| <code>digits</code>   | Number of digits for formatting of numbers                                                                                                            |
| <code>...</code>      | Other arguments (for consistency with the generic)                                                                                                    |

### Value

Plots stopping curves (if specified) and invisibly returns a list of data frames containing the data underlying the stopping curves

### Author(s)

Samuel Pawel

### See Also

[pbf01seq](#)

### Examples

```

n <- seq(25, 150, 25) # sample size per stage
se <- sqrt(2/n) # standard errors per stage
res <- pbf01seq(k1 = 1/10, k0 = 3, se = se, n = n, pm = 0, psd = 1,
               dpm = 0.5, dpsd = 0.1, type = "moment")
plot(res, nullplot = FALSE) # only under design prior
plot(res, nullplot = TRUE) # also plot under null hypothesis
plot(res, zplot = TRUE) # show critical z-values

```

---

|                   |                                             |
|-------------------|---------------------------------------------|
| plot.power.bftest | <i>Plot method for class "power.bftest"</i> |
|-------------------|---------------------------------------------|

---

**Description**

Plot method for class "power.bftest"

**Usage**

```
## S3 method for class 'power.bftest'
plot(
  x,
  nlim = c(2, 500),
  ngrid = 100,
  type = "l",
  plot = TRUE,
  nullplot = TRUE,
  ...
)
```

**Arguments**

|          |                                                                                                                                                          |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| x        | Object of class "power.bftest"                                                                                                                           |
| nlim     | Range of sample sizes over which the power should be computed. Defaults to c(2, 500)                                                                     |
| ngrid    | Number of grid point for which power should be computed. Defaults to 100                                                                                 |
| type     | Type of plot. Defaults to "l" (line-plot)                                                                                                                |
| plot     | Logical indicating whether data should be plotted. If FALSE only the data used for plotting are returned.                                                |
| nullplot | Logical indicating whether a second plot with the power in favor of the null (using a Bayes factor threshold of 1/k) should be created. Defaults to TRUE |
| ...      | Other arguments (for consistency with the generic)                                                                                                       |

**Value**

Plots power curves (if specified) and invisibly returns a list of data frames containing the data underlying the power curves

**Author(s)**

Samuel Pawel

**See Also**

[powerbf01](#), [powerbf01](#), [powernmbf01](#)

## Examples

```
ssd1 <- powerbf01(k = 1/6, power = 0.95, pm = 0, psd = 1/sqrt(2), dpm = 0.5, dpsd = 0)
plot(ssd1, nlim = c(1, 8000))

power1 <- powerbf01(k = 1/2, n = 120, pm = 0, psd = 1/sqrt(2), dpm = 0.5, dpsd = 0)
plot(power1, nlim = c(1, 1000))
```

---

|         |                                                                                 |
|---------|---------------------------------------------------------------------------------|
| pnmbf01 | <i>Cumulative distribution function of the normal moment prior Bayes factor</i> |
|---------|---------------------------------------------------------------------------------|

---

## Description

This function computes the probability of obtaining a normal moment prior Bayes factor ([nmbf01](#)) more extreme than a threshold  $k$  with a specified sample size.

## Usage

```
pnmbf01(k, n, usd, null = 0, psd, dpm, dpsd, lower.tail = TRUE)
```

## Arguments

|                         |                                                                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>k</code>          | Bayes factor threshold                                                                                                                                     |
| <code>n</code>          | Sample size                                                                                                                                                |
| <code>usd</code>        | Unit standard deviation, the (approximate) standard error of the parameter estimate based on $n = 1$ , see details                                         |
| <code>null</code>       | Parameter value under the point null hypothesis. Defaults to 0                                                                                             |
| <code>psd</code>        | Spread of the normal moment prior assigned to the parameter under the alternative in the analysis. The modes of the prior are located at $\pm\sqrt{2}$ psd |
| <code>dpm</code>        | Mean of the normal design prior assigned to the parameter                                                                                                  |
| <code>dpsd</code>       | Standard deviation of the normal design prior assigned to the parameter. Set to 0 to obtain a point prior at the design prior mean                         |
| <code>lower.tail</code> | Logical indicating whether $\Pr(\text{BF}_{01} \leq k)$ (TRUE) or $\Pr(\text{BF}_{01} > k)$ (FALSE) should be computed. Defaults to TRUE                   |

## Details

It is assumed that the standard error of the future parameter estimate is of the form  $\text{se} = \text{usd}/\sqrt{n}$ . For example, for normally distributed data with known standard deviation  $\text{sd}$  and two equally sized groups of size  $n$ , the standard error of an estimated standardized mean difference is  $\text{se} = \text{sd}\sqrt{2/n}$ , so the corresponding unit standard deviation is  $\text{usd} = \text{sd}\sqrt{2}$ . See the vignette for more information.

**Value**

The probability that the Bayes factor is less or greater (depending on the specified `lower.tail`) than the specified threshold `k`

**Author(s)**

Samuel Pawel

**See Also**

[nmbf01](#), [nnmbf01](#), [powernmbf01](#)

**Examples**

```
## point desing prior (psd = 0)
pnmbf01(k = 1/10, n = 200, usd = 2, null = 0, psd = 0.5/sqrt(2), dpm = 0.5, dpsd = 0)

## normal design prior to incorporate parameter uncertainty (psd > 0)
pnmbf01(k = 1/10, n = 200, usd = 2, null = 0, psd = 0.5/sqrt(2), dpm = 0.5, dpsd = 0.25)

## design prior is the null hypothesis (dpm = 0, dpsd = 0)
pnmbf01(k = 10, n = 200, usd = 2, null = 0, psd = 0.5/sqrt(2), dpm = 0, dpsd = 0,
        lower.tail = FALSE)
```

---

powerbf01

*Power and sample size calculations for z-test Bayes factor*

---

**Description**

Compute probability that z-test Bayes factor is smaller than a specified threshold (the power), or determine sample size to obtain a target power.

**Usage**

```
powerbf01(
  n = NULL,
  power = NULL,
  k = 1/10,
  sd = 1,
  null = 0,
  pm,
  psd,
  type = c("two.sample", "one.sample", "paired"),
  dpm = pm,
  dpsd = psd,
  nrange = c(1, 10^5)
)
```

**Arguments**

|        |                                                                                                                                                                                                    |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| n      | Sample size (per group for two-sample tests). Has to be NULL if power is specified. Defaults to NULL                                                                                               |
| power  | Target power. Has to be NULL if n is specified. Defaults to NULL                                                                                                                                   |
| k      | Bayes factor threshold. Defaults to 1/10, Jeffreys' threshold for 'strong evidence' against the null hypothesis                                                                                    |
| sd     | Standard deviation of one observation (for type = "two.sample" or type = "one.sample") or of one difference within a pair of observations (type = "paired"). Is assumed to be known. Defaults to 1 |
| null   | Mean difference under the point null hypothesis. Defaults to 0                                                                                                                                     |
| pm     | Mean of the normal prior assigned to the mean difference under the alternative in the analysis                                                                                                     |
| psd    | Standard deviation of the normal prior assigned to the mean difference under the alternative in the analysis. Set to 0 to obtain a point prior at the prior mean                                   |
| type   | The type of test. One of "two.sample", "one.sample", "paired". Defaults to "two.sample"                                                                                                            |
| dpm    | Mean of the normal design prior assigned to the mean difference. Defaults to the same value as the analysis prior pm                                                                               |
| dpsd   | Standard deviation of the normal design prior assigned to the mean difference. Defaults to the same value as the analysis prior psd                                                                |
| nrange | Sample size search range over which numerical search is performed (only taken into account when n is NULL). Defaults to c(1, 10^5)                                                                 |

**Details**

This function provides a similar interface as `stats::power.t.test`. It also assumes that the data are continuous and that the parameter of interest is either a mean or a (standardized) mean difference. For some users, the low-level functions [nbf01](#) (to directly compute the sample size for a fixed power) and [pbf01](#) (to directly compute the power for a fixed sample size) may also be useful because they can be used for other data and parameter types.

**Value**

Object of class "power.bftest", a list of the arguments (including the computed one) augmented with method and note elements

**Note**

A warning message will be displayed in case that the specified target power is not achievable under the specified analysis and design priors.

**Author(s)**

Samuel Pawel



**See Also**

[plot.power.bftest](#), [nbf01](#), [pbf01](#), [bf01](#)

**Examples**

```
## determine power
powerbinbf01(n = 100, pm = 0, psd = 1, dpm = 0.5, dpsd = 0)

## determine sample size
powerbinbf01(power = 0.99, pm = 0, psd = 1, dpm = 0.5, dpsd = 0)
```

---

powerbinbf01

*Power and sample size calculations for binomial Bayes factor*

---

**Description**

Compute probability that binomial Bayes factor ([binbf01](#)) is smaller than a specified threshold (the power), or determine sample size to obtain a target power.

**Usage**

```
powerbinbf01(
  n = NULL,
  power = NULL,
  k = 1/10,
  p0 = 0.5,
  type = c("point", "direction"),
  a = 1,
  b = 1,
  dp = NA,
  da = a,
  db = b,
  dl = 0,
  du = 1,
  nrange = c(1, 10^4)
)
```

**Arguments**

|       |                                                                                                                 |
|-------|-----------------------------------------------------------------------------------------------------------------|
| n     | Sample size. Has to be NULL if power is specified. Defaults to NULL                                             |
| power | Target power. Has to be NULL if n is specified. Defaults to NULL                                                |
| k     | Bayes factor threshold. Defaults to 1/10, Jeffreys' threshold for 'strong evidence' against the null hypothesis |
| p0    | Tested binomial proportion. Defaults to 0.5                                                                     |
| type  | Type of test. Can be "point" or "directional". Defaults to "point"                                              |

|        |                                                                                                                                                                                        |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a      | Number of successes parameter of the beta analysis prior distribution. Defaults to 1                                                                                                   |
| b      | Number of failures parameter of the beta analysis prior distribution. Defaults to 1                                                                                                    |
| dp     | Fixed binomial proportion assumed for the power calculation. Set to NA to use a truncated beta design prior instead (specified via the da, db, dl, and du arguments). Defaults to NA   |
| da     | Number of successes parameter of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to the same value a as specified for the analysis prior |
| db     | Number of failures parameter of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to the same value b as specified for the analysis prior  |
| dl     | Lower truncation limit of of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to 0                                                        |
| du     | Upper truncation limit of of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to 1                                                        |
| nrange | Sample size search range over which numerical search is performed (only taken into account when n is NULL). Defaults to c(1, 10^3)                                                     |

### Details

This function provides a similar interface as `stats::power.prop.test`. For some users, the low-level functions [nbinbf01](#) (to directly compute the sample size for a fixed power) and [pbinbf01](#) (to directly compute the power for a fixed sample size) may also be useful.

### Value

Object of class "power.bftest", a list of the arguments (including the computed one) augmented with method and note elements

### Author(s)

Samuel Pawel

### See Also

[plot.power.bftest](#), [pbinbf01](#), [nbinbf01](#), [binbf01](#)

### Examples

```
## determine sample size
(nres <- powerbinbf01(power = 0.8, p0 = 0.2, type = "direction", dl = 0.2))
## Not run:
plot(nres, nlim = c(1, 250), ngrid = 250, type = "s")

## End(Not run)
```

```
## determine power
(powres <- powerbinbf01(n = 100, type = "point"))
## Not run:
plot(powres)

## End(Not run)
```

---

|             |                                                                                |
|-------------|--------------------------------------------------------------------------------|
| powernmbf01 | <i>Power and sample size calculations for normal moment prior Bayes factor</i> |
|-------------|--------------------------------------------------------------------------------|

---

## Description

Compute probability that normal moment prior Bayes factor is smaller than a specified threshold (the power), or determine sample size to obtain a target power.

## Usage

```
powernmbf01(
  n = NULL,
  power = NULL,
  k = 1/10,
  sd = 1,
  null = 0,
  psd,
  type = c("two.sample", "one.sample", "paired"),
  dpm,
  dpsd,
  nrange = c(1, 10^5)
)
```

## Arguments

|       |                                                                                                                                                                                                    |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| n     | Sample size (per group for two-sample tests). Has to be NULL if power is specified. Defaults to NULL                                                                                               |
| power | Target power. Has to be NULL if n is specified. Defaults to NULL                                                                                                                                   |
| k     | Bayes factor threshold. Defaults to 1/10, Jeffreys' threshold for 'strong evidence' against the null hypothesis                                                                                    |
| sd    | Standard deviation of one observation (for type = "two.sample" or type = "one.sample") or of one difference within a pair of observations (type = "paired"). Is assumed to be known. Defaults to 1 |
| null  | Parameter value under the point null hypothesis. Defaults to 0                                                                                                                                     |
| psd   | Spread of the normal moment prior assigned to the parameter under the alternative in the analysis. The modes of the prior are located at $\pm\sqrt{2}$ psd                                         |

|        |                                                                                                                                    |
|--------|------------------------------------------------------------------------------------------------------------------------------------|
| type   | The type of test. One of "two.sample", "one.sample", "paired". Defaults to "two.sample"                                            |
| dpm    | Mean of the normal design prior assigned to the parameter                                                                          |
| dpsd   | Standard deviation of the normal design prior assigned to the parameter. Set to 0 to obtain a point prior at the design prior mean |
| nrange | Sample size search range over which numerical search is performed (only taken into account when n is NULL). Defaults to c(1, 10^5) |

### Details

This function provides a similar interface as `stats::power.t.test`. It also assumes that the data are continuous and that the parameter of interest is either a mean or a (standardized) mean difference. For some users, the low-level functions [nnmbf01](#) (to directly compute the sample size for a fixed power) and [pnmbf01](#) (to directly compute the power for a fixed sample size) may also be useful because they can be used for other data and parameter types.

### Value

Object of class "power.bftest", a list of the arguments (including the computed one) augmented with method and note elements

### Author(s)

Samuel Pawel

### See Also

[plot.power.bftest](#), [nnmbf01](#), [pnmbf01](#), [nmbf01](#)

### Examples

```
## determine power
powernnmbf01(n = 100, psd = 1, dpm = 0.5, dpsd = 0)

## determine sample size
powernnmbf01(power = 0.99, psd = 1, dpm = 0.5, dpsd = 0)
```

---

powertbf01

*Power and sample size calculations for t-test Bayes factor*

---

### Description

Compute probability that *t*-test Bayes factor is smaller than a specified threshold (the power), or determine sample size to obtain a target power.

**Usage**

```
powerxbf01(
  n = NULL,
  power = NULL,
  k = 1/10,
  null = 0,
  plocation = 0,
  pscale = 1/sqrt(2),
  pdf = 1,
  type = c("two.sample", "one.sample", "paired"),
  alternative = c("two.sided", "less", "greater"),
  dpm = plocation,
  dpsd = pscale,
  nrange = c(2, 10^4)
)
```

**Arguments**

|             |                                                                                                                                                                                                                                                                                                                |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| n           | Sample size (per group)                                                                                                                                                                                                                                                                                        |
| power       | Target power. Has to be NULL if n is specified. Defaults to NULL                                                                                                                                                                                                                                               |
| k           | Bayes factor threshold. Defaults to 1/10, Jeffreys' threshold for 'strong evidence' against the null hypothesis                                                                                                                                                                                                |
| null        | Standardized mean difference under the point null hypothesis. Defaults to 0                                                                                                                                                                                                                                    |
| plocation   | <i>t</i> prior location. Defaults to 0                                                                                                                                                                                                                                                                         |
| pscale      | <i>t</i> prior scale. Defaults to 1/sqrt(2)                                                                                                                                                                                                                                                                    |
| pdf         | <i>t</i> prior degrees of freedom. Defaults to 1 (a Cauchy prior)                                                                                                                                                                                                                                              |
| type        | Type of <i>t</i> -test. Can be "two.sample" (default), "one.sample", or "paired"                                                                                                                                                                                                                               |
| alternative | Direction of the test. Can be either "two.sided" (default), "less", or "greater". The latter two truncate the analysis prior to negative and positive effects, respectively. If set to "less" or "greater", the power is only computed based on data with effect estimates in the direction of the alternative |
| dpm         | Mean of the normal design prior assigned to the standardized mean difference. Defaults to the analysis prior location                                                                                                                                                                                          |
| dpsd        | Standard deviation of the normal design prior assigned to the standardized mean difference. Set to 0 to obtain a point prior at the design prior mean. Defaults to the analysis prior scale                                                                                                                    |
| nrange      | Sample size search range over which numerical search is performed (only taken into account when n is NULL). Defaults to c(2, 10^4)                                                                                                                                                                             |

**Details**

This function provides a similar interface as `stats::power.t.test`. For some users, the low-level functions [ntbf01](#) (to directly compute the sample size for a fixed power) and [ptbf01](#) (to directly compute the power for a fixed sample size) may also be useful.

**Value**

Object of class "power.bftest", a list of the arguments (including the computed one) augmented with method and note elements

**Author(s)**

Samuel Pawel

**See Also**

[plot.power.bftest](#), [ptbf01](#), [ntbf01](#), [tbf01](#)

**Examples**

```
## determine power
powerbf01(n = 146, k = 1/6, dpm = 0.5, dps = 0, alternative = "greater")

## determine sample size
powerbf01(power = 0.95, k = 1/6, dpm = 0.5, dps = 0, alternative = "greater")
```

---

```
print.bfseqdesign      Print method for class "bfseqdesign"
```

---

**Description**

Print method for class "bfseqdesign"

**Usage**

```
## S3 method for class 'bfseqdesign'
print(x, digits = max(3L, getOption("digits") - 3L), ...)
```

**Arguments**

|        |                                                    |
|--------|----------------------------------------------------|
| x      | Object of class "bfseqdesign"                      |
| digits | Number of digits for formatting of numbers         |
| ...    | Other arguments (for consistency with the generic) |

**Value**

Prints text summary in the console and invisibly returns the "bfseqdesign" object

**Author(s)**

Samuel Pawel

**See Also**[pbf01seq](#)**Examples**

```
n <- seq(50, 300, 50) # sample size per stage
se <- sqrt(2/n) # standard errors per stage
res <- pbf01seq(k1 = 1/10, k0 = 5, se = se, n = n, pm = 0, psd = 1,
               dpm = 0.5, dpsd = 0.1, type = "normal")
res
```

---

|                    |                                              |
|--------------------|----------------------------------------------|
| print.power.bftest | <i>Print method for class "power.bftest"</i> |
|--------------------|----------------------------------------------|

---

**Description**

Print method for class "power.bftest"

**Usage**

```
## S3 method for class 'power.bftest'
print(x, digits = getOption("digits"), ...)
```

**Arguments**

|        |                                                    |
|--------|----------------------------------------------------|
| x      | Object of class "power.bftest"                     |
| digits | Number of digits for formatting of numbers         |
| ...    | Other arguments (for consistency with the generic) |

**Value**

Prints text summary in the console and invisibly returns the "power.bftest" object

**Note**

Function adapted from stats:::print.power.htest written by Peter Dalgaard

**Author(s)**

Samuel Pawel

**See Also**[powerbf01](#)

## Examples

```
powerbf01(power = 0.95, pm = 0, psd = 1, dpm = 0.5, dpsd = 0)
powerbf01(power = 0.95, pm = 0, psd = 1, dpm = 0.5, dpsd = 0, type = "one.sample")
powerbf01(power = 0.95, pm = 0, psd = 1, dpm = 0.5, dpsd = 0, type = "paired")
powerbf01(power = 0.95, pm = 1, psd = 0, dpm = 0.8, dpsd = 0, type = "paired")
```

---

ptbf01

---

*Cumulative distribution function of the  $t$ -test Bayes factor*


---

## Description

This function computes the probability of obtaining a  $t$ -test Bayes factor ([tbf01](#)) more extreme than a threshold  $k$  with a specified sample size.

## Usage

```
ptbf01(
  k,
  n,
  n1 = n,
  n2 = n,
  null = 0,
  plocation = 0,
  pscale = 1/sqrt(2),
  pdf = 1,
  dpm = plocation,
  dpsd = pscale,
  type = c("two.sample", "one.sample", "paired"),
  alternative = c("two.sided", "less", "greater"),
  lower.tail = TRUE,
  drange = "adaptive",
  ...
)
```

## Arguments

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <code>k</code>         | Bayes factor threshold                                                                   |
| <code>n</code>         | Sample size (per group)                                                                  |
| <code>n1</code>        | Sample size in group 1 (only required for two-sample $t$ -test with unequal group sizes) |
| <code>n2</code>        | Sample size in group 2 (only required for two-sample $t$ -test with unequal group sizes) |
| <code>null</code>      | Standardized mean difference under the point null hypothesis. Defaults to 0              |
| <code>plocation</code> | $t$ prior location. Defaults to 0                                                        |



|             |                                                                                                                                                                                                                                                                                                                |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pscale      | $t$ prior scale. Defaults to $1/\sqrt{2}$                                                                                                                                                                                                                                                                      |
| pdf         | $t$ prior degrees of freedom. Defaults to 1 (a Cauchy prior)                                                                                                                                                                                                                                                   |
| dpm         | Mean of the normal design prior assigned to the standardized mean difference. Defaults to the analysis prior location                                                                                                                                                                                          |
| dpsd        | Standard deviation of the normal design prior assigned to the standardized mean difference. Set to 0 to obtain a point prior at the design prior mean. Defaults to the analysis prior scale                                                                                                                    |
| type        | Type of $t$ -test. Can be "two.sample" (default), "one.sample", or "paired"                                                                                                                                                                                                                                    |
| alternative | Direction of the test. Can be either "two.sided" (default), "less", or "greater". The latter two truncate the analysis prior to negative and positive effects, respectively. If set to "less" or "greater", the power is only computed based on data with effect estimates in the direction of the alternative |
| lower.tail  | Logical indicating whether $\Pr(\text{BF}_{01} \leq k)$ (TRUE) or $\Pr(\text{BF}_{01} > k)$ (FALSE) should be computed. Defaults to TRUE                                                                                                                                                                       |
| drange      | Standardized mean difference search range over which the critical values are searched for. Can be either set to a numerical range or to "adaptive" (default) which determines the range in an adaptive way from the other input parameters                                                                     |
| ...         | Other arguments passed to <code>stats::uniroot</code>                                                                                                                                                                                                                                                          |

**Value**

The probability that the Bayes factor is less or greater (depending on the specified `lower.tail`) than the specified threshold `k`

**Author(s)**

Samuel Pawel

**See Also**

[tbf01](#), [ntbf01](#), [powertbf01](#)

**Examples**

```
## example from Schönbrodt and Wagenmakers (2018, p. 135)
ptbf01(k = 1/6, n = 146, dpm = 0.5, dpsd = 0, alternative = "greater")
ptbf01(k = 6, n = 146, dpm = 0, dpsd = 0, alternative = "greater",
      lower.tail = FALSE)

## two-sided
ptbf01(k = 1/6, n = 146, dpm = 0.5, dpsd = 0)
ptbf01(k = 6, n = 146, dpm = 0, dpsd = 0, lower.tail = FALSE)

## one-sample test
ptbf01(k = 1/6, n = 146, dpm = 0.5, dpsd = 0, alternative = "greater",
      type = "one.sample")
```

ptbf01seq

*Sequential T-Test Bayes Factor Design***Description**

Computes cumulative probabilities of observing  $t$ -test Bayes factors that provide evidence for the null hypothesis  $H_0$ , the alternative hypothesis  $H_1$ , or remain inconclusive in a sequential design. Also computes the expected sample size.

**Usage**

```
ptbf01seq(
  k1,
  k0 = 1/k1,
  n,
  n1 = n,
  n2 = n,
  plocation = 0,
  pscale = 1/sqrt(2),
  pdf = 1,
  dpm = plocation,
  dpsd = pscale,
  type = c("two.sample", "one.sample", "paired"),
  alternative = c("two.sided", "less", "greater"),
  strict = TRUE,
  drange = "adaptive",
  ...
)
```

**Arguments**

|           |                                                                                                                       |
|-----------|-----------------------------------------------------------------------------------------------------------------------|
| k1        | Bayes factor threshold in favor of $H_1$ (i.e., $\text{BF}_{01} \leq k1 < 1$ implies evidence for $H_1$ )             |
| k0        | Bayes factor threshold in favor of $H_0$ (i.e., $\text{BF}_{01} \geq k0 > 1$ implies evidence for $H_1$ )             |
| n         | Sample size (per group)                                                                                               |
| n1        | Sample size in group 1 (only required for two-sample $t$ -test with unequal group sizes)                              |
| n2        | Sample size in group 2 (only required for two-sample $t$ -test with unequal group sizes)                              |
| plocation | $t$ prior location. Defaults to 0                                                                                     |
| pscale    | $t$ prior scale. Defaults to $1/\sqrt{2}$                                                                             |
| pdf       | $t$ prior degrees of freedom. Defaults to 1 (a Cauchy prior)                                                          |
| dpm       | Mean of the normal design prior assigned to the standardized mean difference. Defaults to the analysis prior location |

|             |                                                                                                                                                                                                                                                                                                                                                                     |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dpsd        | Standard deviation of the normal design prior assigned to the standardized mean difference. Set to 0 to obtain a point prior at the design prior mean. Defaults to the analysis prior scale                                                                                                                                                                         |
| type        | Type of $t$ -test. Can be "two.sample" (default), "one.sample", or "paired"                                                                                                                                                                                                                                                                                         |
| alternative | Direction of the test. Can be either "two.sided" (default), "less", or "greater". The latter two truncate the analysis prior to negative and positive effects, respectively. If set to "less" or "greater", the power is only computed based on data with effect estimates in the direction of the alternative                                                      |
| strict      | Logical. If TRUE and there are more than two critical values per stage, integrate over all possible region combinations (slow but exact). If FALSE, only integrates over the main regions where the sign of the $z$ -statistics does not change across stages (faster, recommended when many interim analyses, e.g., more than 10, are performed). Defaults to TRUE |
| drange      | Standardized mean difference search range over which the critical values are searched for. Can be either set to a numerical range or to "adaptive" (default) which determines the range in an adaptive way from the other input parameters                                                                                                                          |
| ...         | Additional arguments passed to <code>mvtnorm::lpmvnorm</code>                                                                                                                                                                                                                                                                                                       |

## Details

The function constructs per-stage integration regions for cumulative  $z$ -statistics based on the Bayes factor thresholds  $k_1$  and  $k_0$ , then computes the probability of these regions under a predictive distribution defined by the asymptotic variance of the  $t$ -statistic and the normal design prior with `dpm` and `dpsd`. Integration is performed via `mvtnorm::lpmvnorm`.

## Value

An object of class "bfseqdesign", which is a list containing the input arguments, the critical  $z$ -values, the expected sample size, the cumulative probabilities of stopping for  $H_1$  and  $H_0$  by each stage, and the cumulative probabilities of remaining inconclusive by each stage.

## Author(s)

Samuel Pawel

## Examples

```
## similar to example from Schönbrodt and Wagenmakers (2018, p. 138)
k0 <- 6
k1 <- 1/30
dpm <- 0.5
dpsd <- 0.1
plocation <- 0
pscale <- 1/sqrt(2)
pdf <- 1
type <- "two.sample"
alternative <- "greater"
n <- seq(40, 100, 10) # sample size (per group) per stage
res <- ptbf01seq(k1 = k1, k0 = k0, n = n, plocation = plocation,
```

```

        pscale = pscale, pdf = pdf, dpm = dpm, dpsd = dpsd,
        alternative = alternative, type = type)
res
plot(res) # show stopping probabilities
plot(res, zplot = TRUE) # show critical z-values

```

---

tbf01

*t-test Bayes factor*


---

## Description

This function computes the Bayes factor that forms the basis of the informed Bayesian *t*-test from Gronau et al. (2020). The Bayes factor quantifies the evidence that the data provide for the null hypothesis that the standardized mean difference (SMD) is zero against the alternative that the SMD is non-zero. A location-scale *t*-distribution is assumed for the SMD under the alternative hypothesis. The Jeffreys-Zellner-Siow (JZS) Bayes factor (Rouder et al., 2009) is obtained as a special case by setting the location of the prior to zero and the prior degrees of freedom to one, which is the default.

The data are summarized by *t*-statistics and sample sizes. The following types of *t*-statistics are accepted:

- Two-sample *t*-test where the SMD represents the standardized mean difference between two group means (assuming equal variances in both groups)
- One-sample *t*-test where the SMD represents the standardized mean difference to the null value
- Paired *t*-test where the SMD represents the standardized mean change score

## Usage

```

tbf01(
  t,
  n,
  n1 = n,
  n2 = n,
  plocation = 0,
  pscale = 1/sqrt(2),
  pdf = 1,
  type = c("two.sample", "one.sample", "paired"),
  alternative = c("two.sided", "less", "greater"),
  log = FALSE,
  ...
)

```

**Arguments**

|             |                                                                                                                                                                              |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| t           | <i>t</i> -statistic                                                                                                                                                          |
| n           | Sample size (per group)                                                                                                                                                      |
| n1          | Sample size in group 1 (only required for two-sample <i>t</i> -test with unequal group sizes)                                                                                |
| n2          | Sample size in group 2 (only required for two-sample <i>t</i> -test with unequal group sizes)                                                                                |
| plocation   | <i>t</i> prior location. Defaults to 0                                                                                                                                       |
| pscale      | <i>t</i> prior scale. Defaults to 1/sqrt(2)                                                                                                                                  |
| pdf         | <i>t</i> prior degrees of freedom. Defaults to 1 (a Cauchy prior)                                                                                                            |
| type        | Type of <i>t</i> -test. Can be "two.sample" (default), "one.sample", or "paired"                                                                                             |
| alternative | Direction of the test. Can be either "two.sided" (default), "less", or "greater". The latter two truncate the analysis prior to negative and positive effects, respectively. |
| log         | Logical indicating whether the natural logarithm of the Bayes factor should be returned. Defaults to FALSE                                                                   |
| ...         | Additional arguments passed to <code>stats::integrate</code>                                                                                                                 |

**Details**

The Bayes factor is implemented as in equation (5) in Gronau et al. (2020), and using suitable truncation in case of one-sided alternatives. Integration is performed numerically with `stats::integrate`.

**Value**

Bayes factor in favor of the null hypothesis over the alternative ( $BF_{01} > 1$  indicates evidence for the null hypothesis, whereas  $BF_{01} < 1$  indicates evidence for the alternative)

**Author(s)**

Samuel Pawel

**References**

Rouder, J. N., Speckman, P. L., Sun, D., Morey, R. D., Iverson, G. (2009). Bayesian *t* tests for accepting and rejecting the null hypothesis. *Psychonomic Bulletin & Review*, 16(2):225-237. [doi:10.3758/PBR.16.2.225](https://doi.org/10.3758/PBR.16.2.225)

Gronau, Q. F., Ly., A., Wagenmakers, E.J. (2020). Informed Bayesian *t*-Tests. *The American Statistician*, 74(2):137-143. [doi:10.1080/00031305.2018.1562983](https://doi.org/10.1080/00031305.2018.1562983)

**See Also**

[powerbtf01](#), [ptbf01](#), [ntbf01](#)

**Examples**

```
## analyses from Rouder et al. (2009):
## values from Table 1
tbfb01(t = c(0.69, 3.20), n = 100, pscale = 1, type = "one.sample")
## examples from p. 232
tbfb01(t = c(2.24, 2.03), n = 80, pscale = 1, type = "one.sample")

## analyses from Gronau et al. (2020) section 3.2:
## informed prior
tbfb01(t = -0.90, n1 = 53, n2 = 57, plocation = 0.350, pscale = 0.102, pdf = 3,
       alternative = "greater", type = "two.sample")
## default (one-sided) prior
tbfb01(t = -0.90, n1 = 53, n2 = 57, plocation = 0, pscale = 1/sqrt(2), pdf = 1,
       alternative = "greater", type = "two.sample")
```

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